SDITO -LAB

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Assignment NO : 09

Problem Statement : Deploy a Project from GitHub to AWS EC2 Instance

Aim

To launch an EC2 instance, connect it via SSH, install required software, clone a project from GitHub, and deploy it so it is accessible via Public IPv4 address.

Requirements

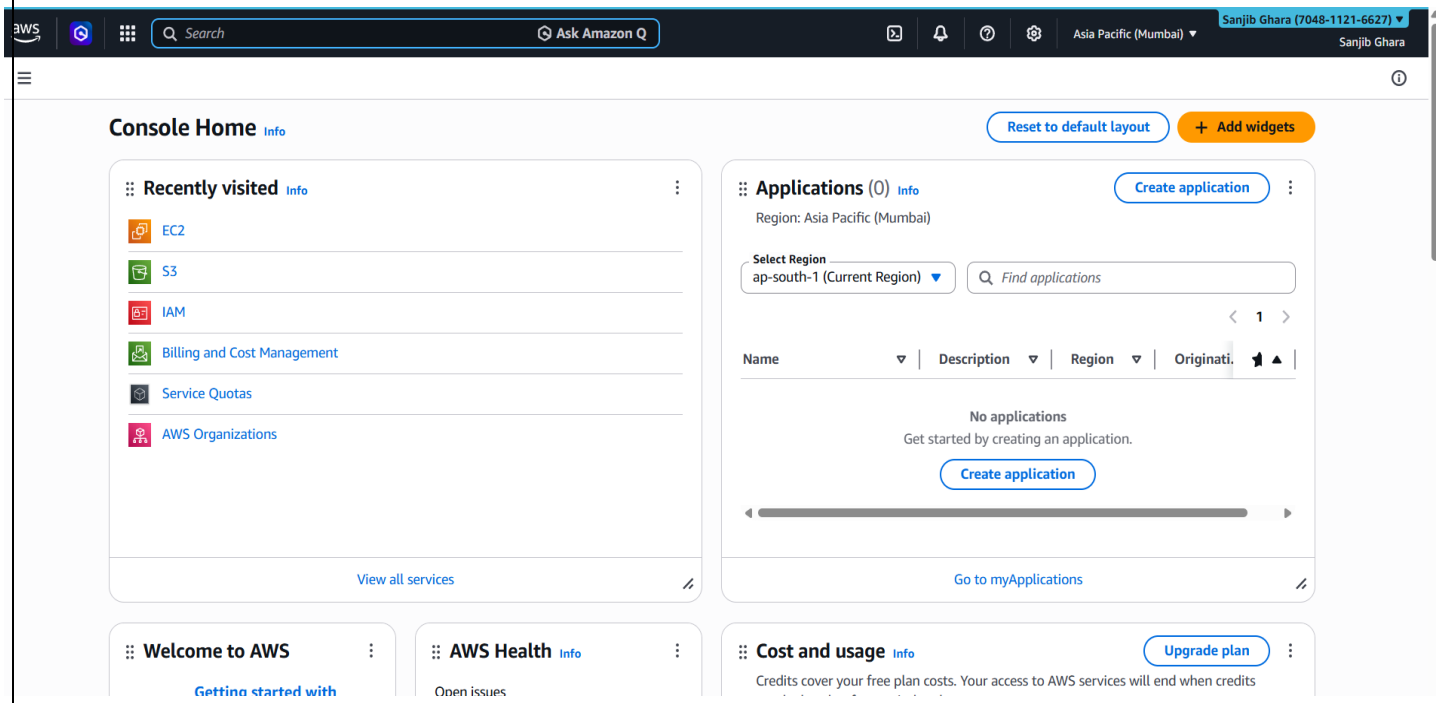
- AWS Account
- EC2 Ubuntu Instance
- Key Pair (.pem)
- GitHub Repository
- Node.js Project
- Security Group with required ports open

Solution:

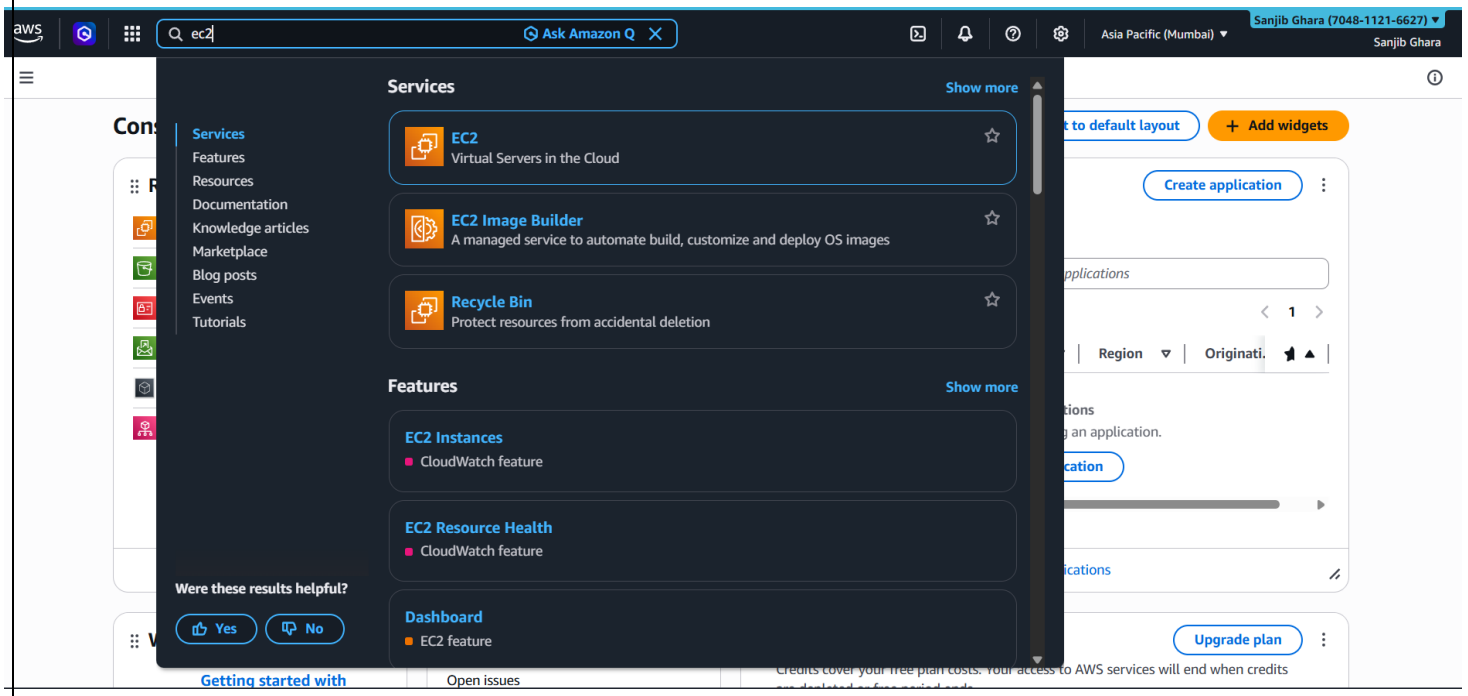
PHASE 1 – Launch EC2 Instance

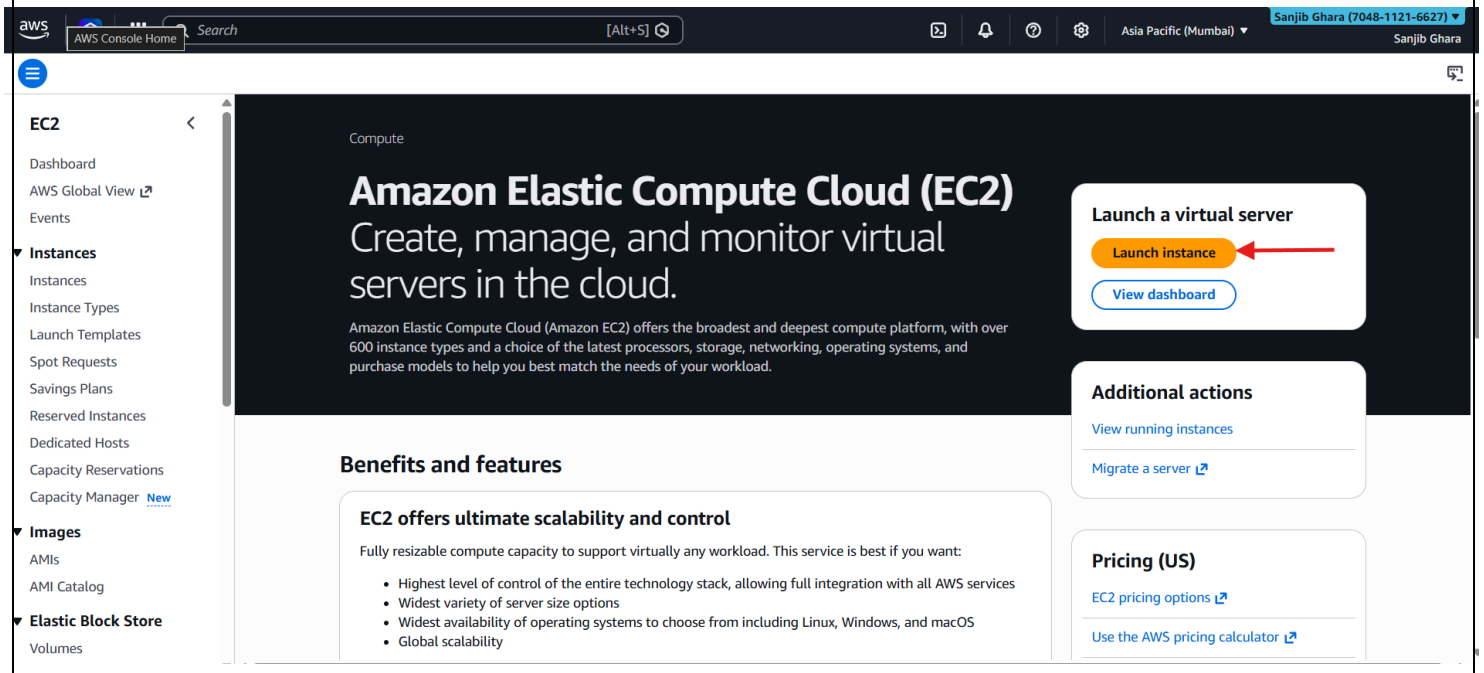
Step 1: Launch an EC2 Instance

1. Login to AWS Management Console.

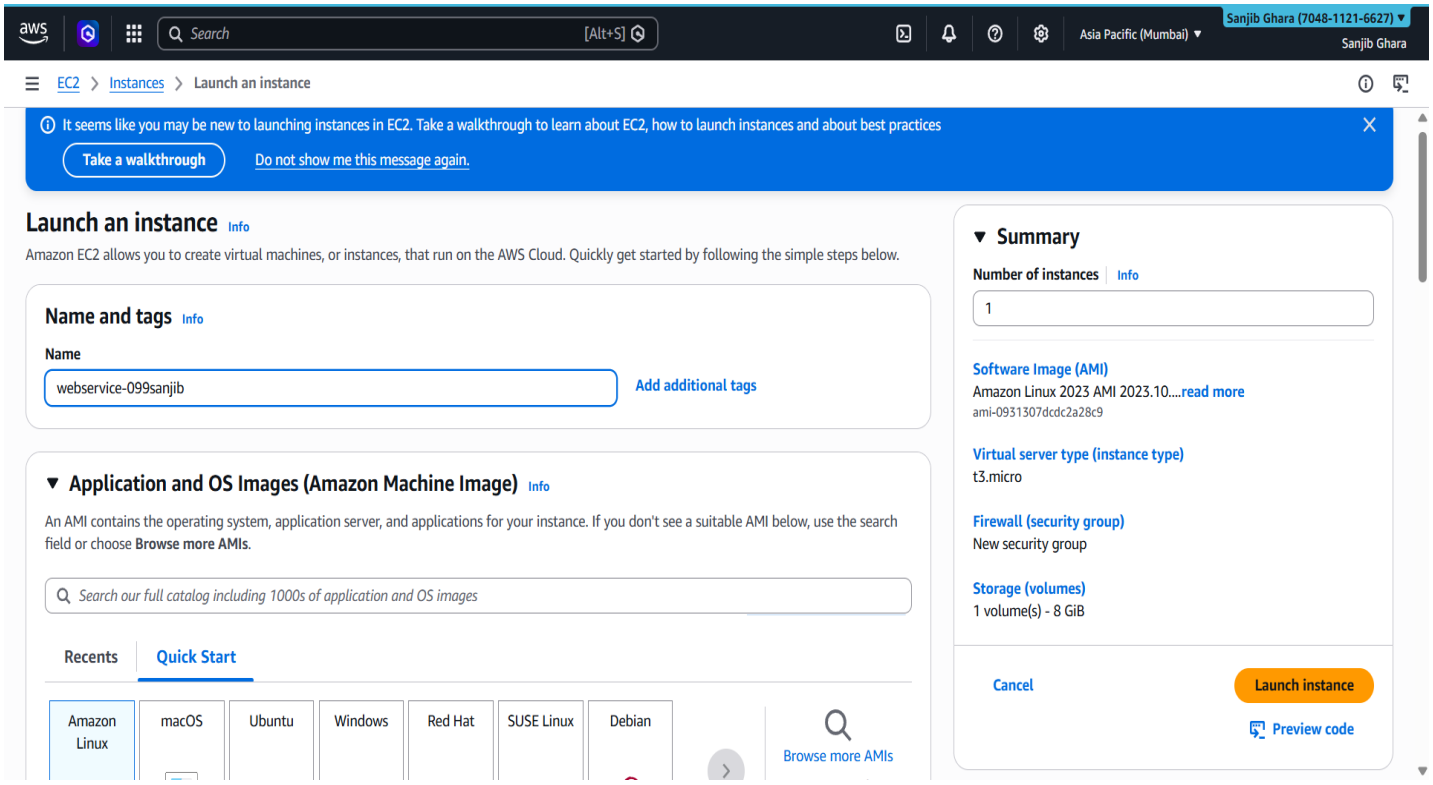


2. Search for EC2 and open the service.

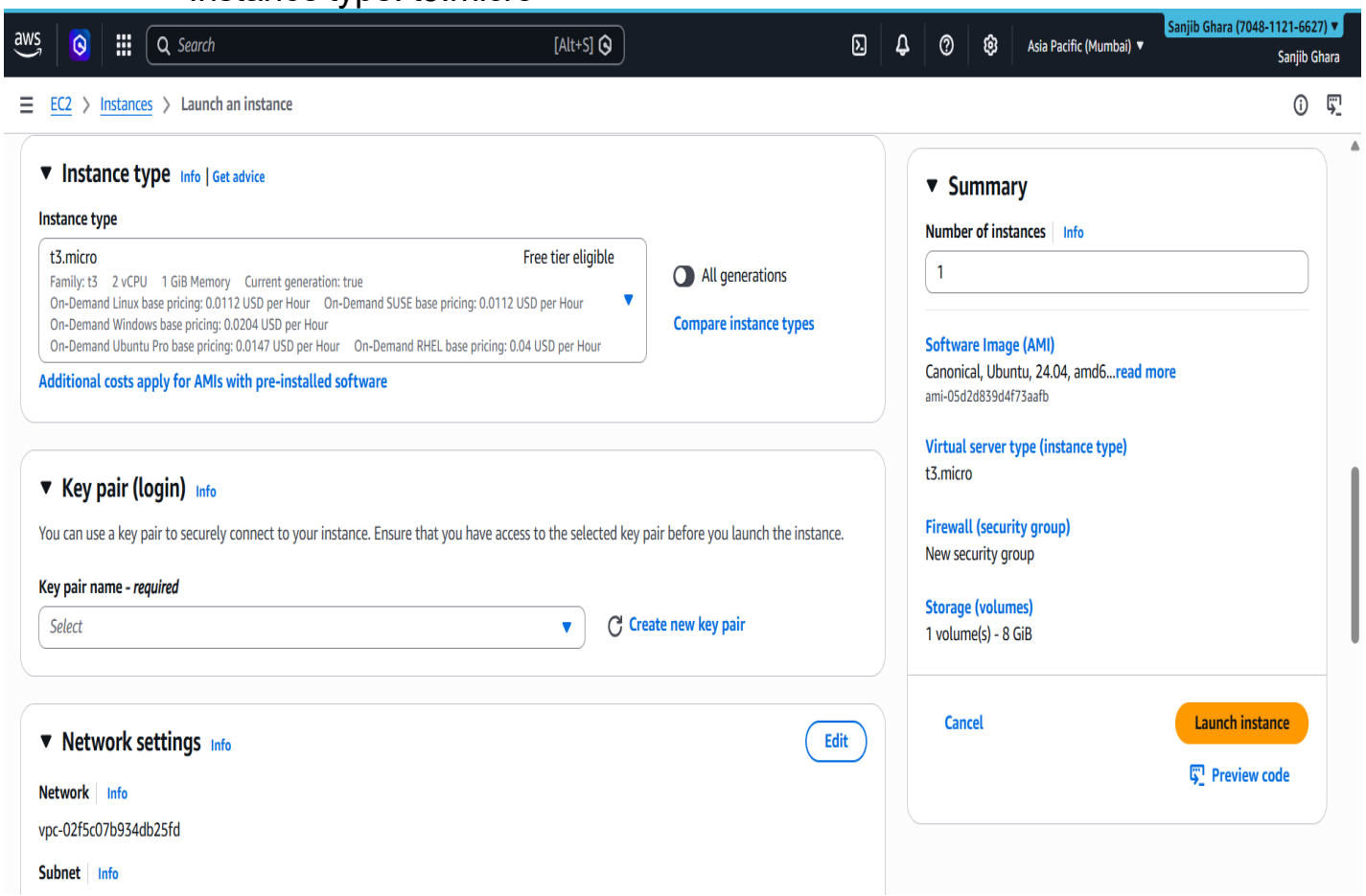




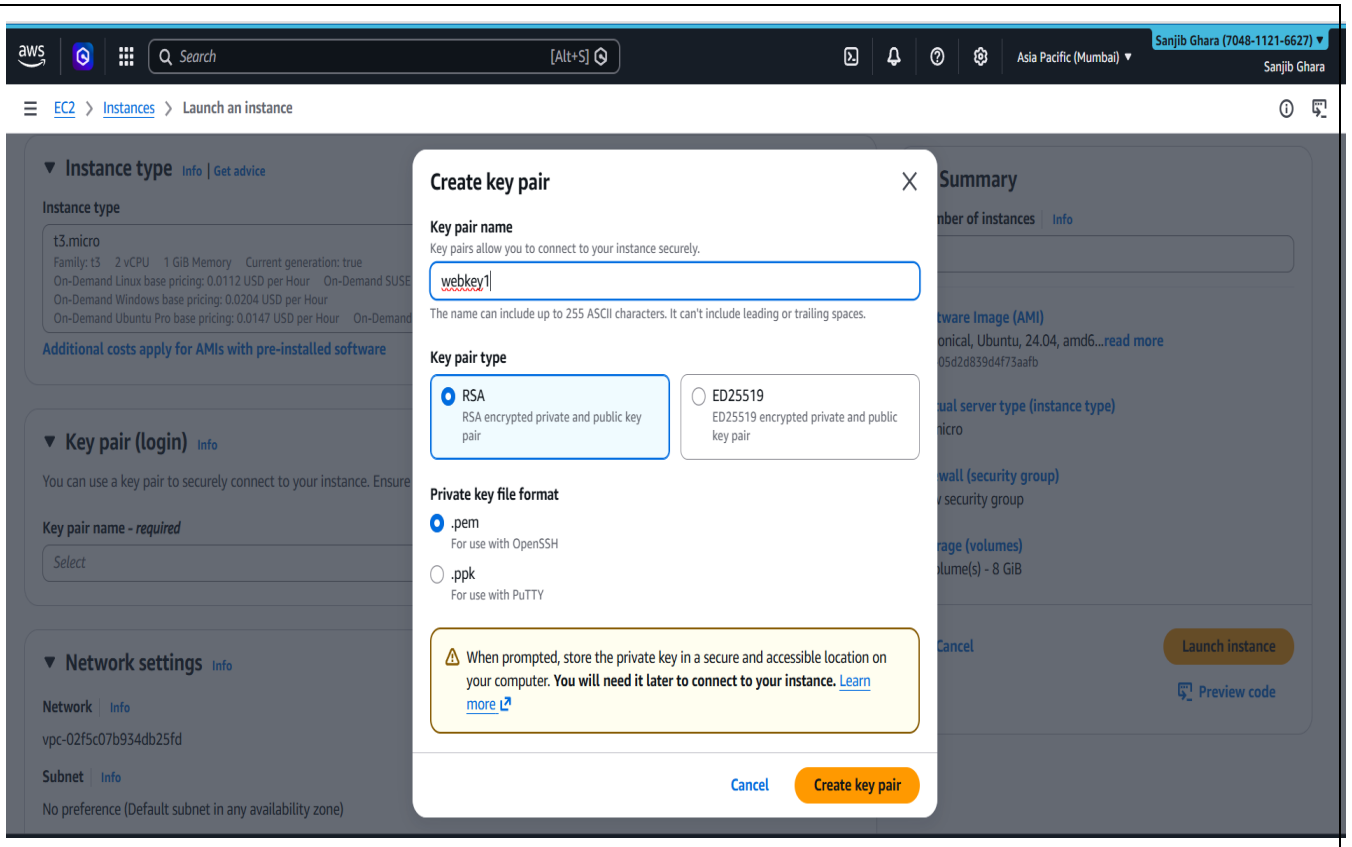
3. Click Launch instance.
   <u>Instance Configuration</u>
4. Enter:
   - Name: WebService-099sanjib
5. Select:
   - AMI: Ubuntu (Free Tier Eligible)



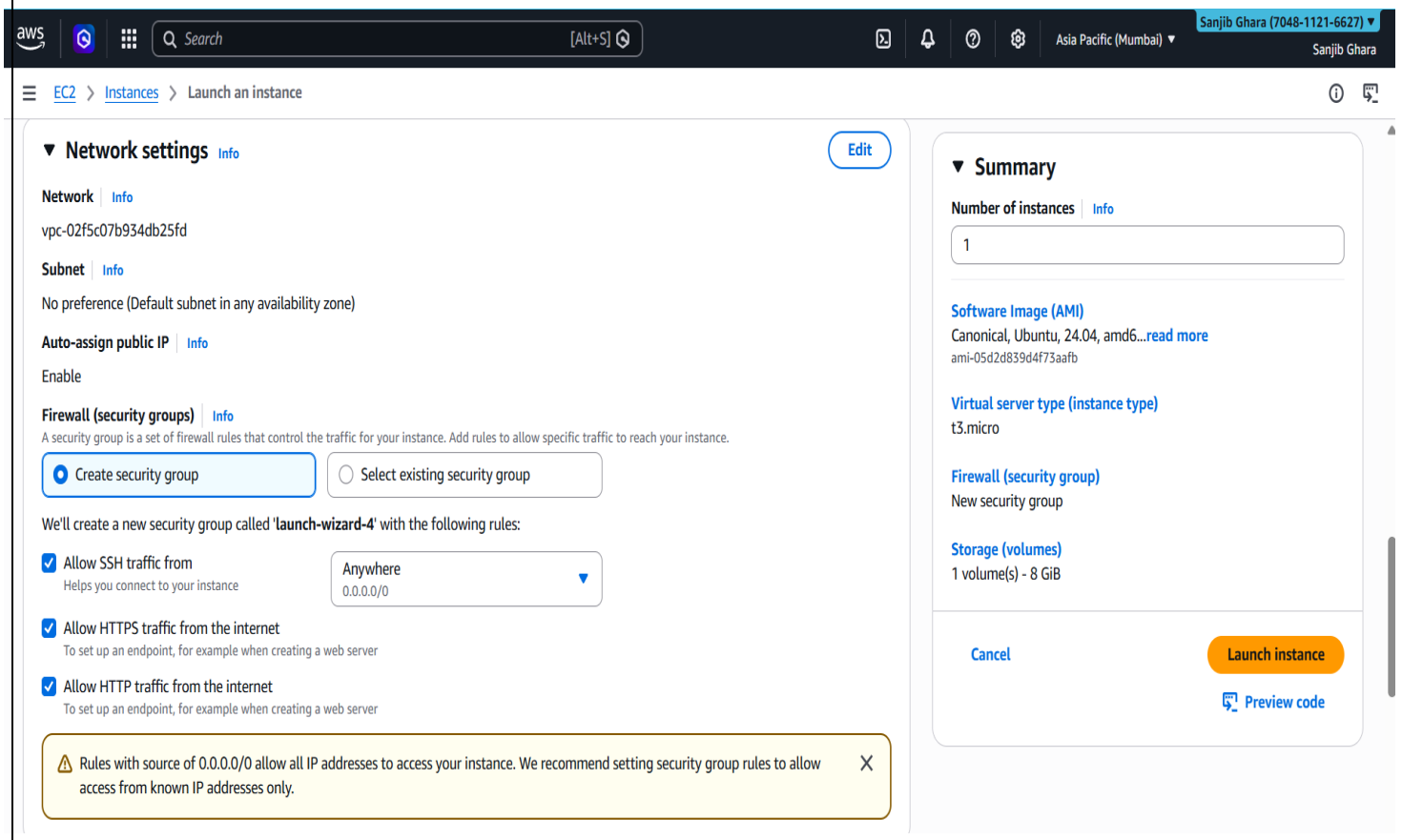
6. Choose:
    - Instance type: t3.micro



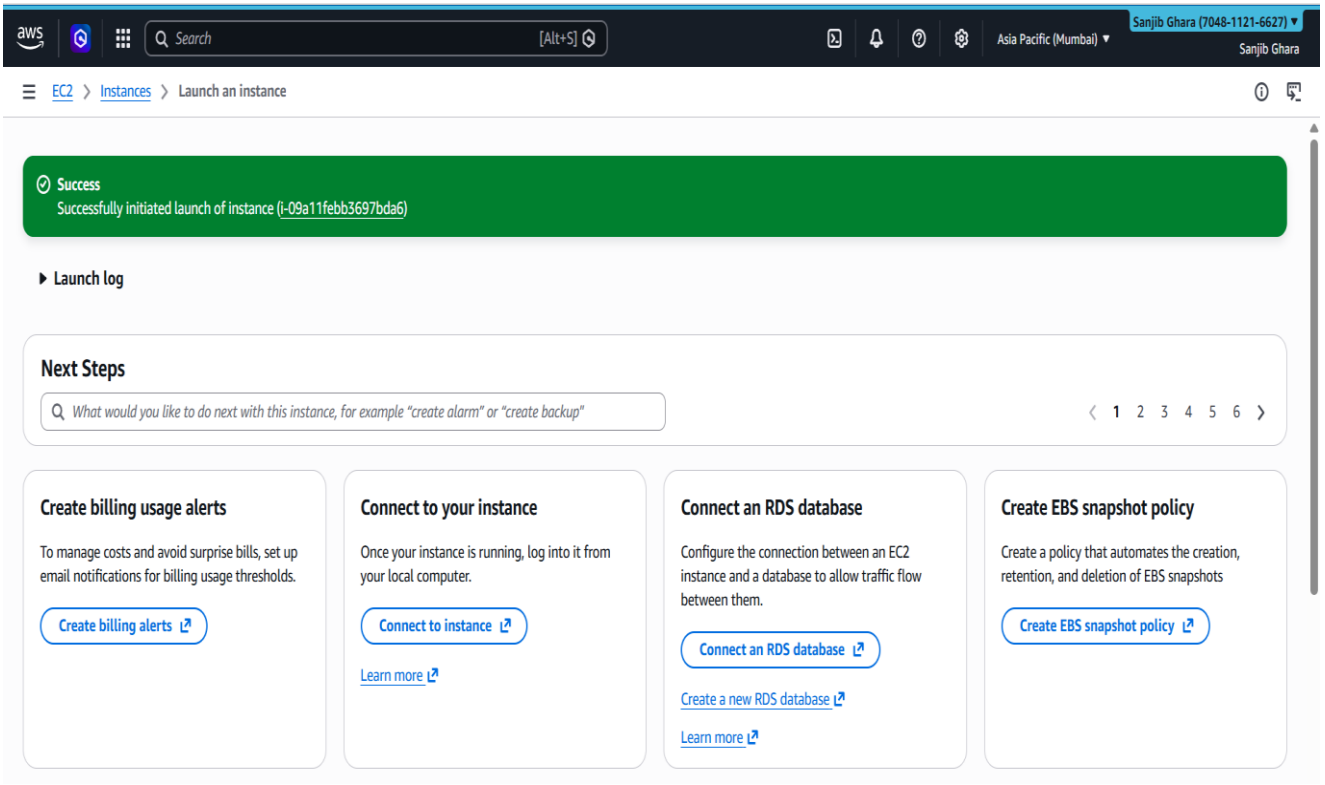
7. Click Create new key pair:
    - Name: Webkey1
    - Type: RSA
    - Format: .pem



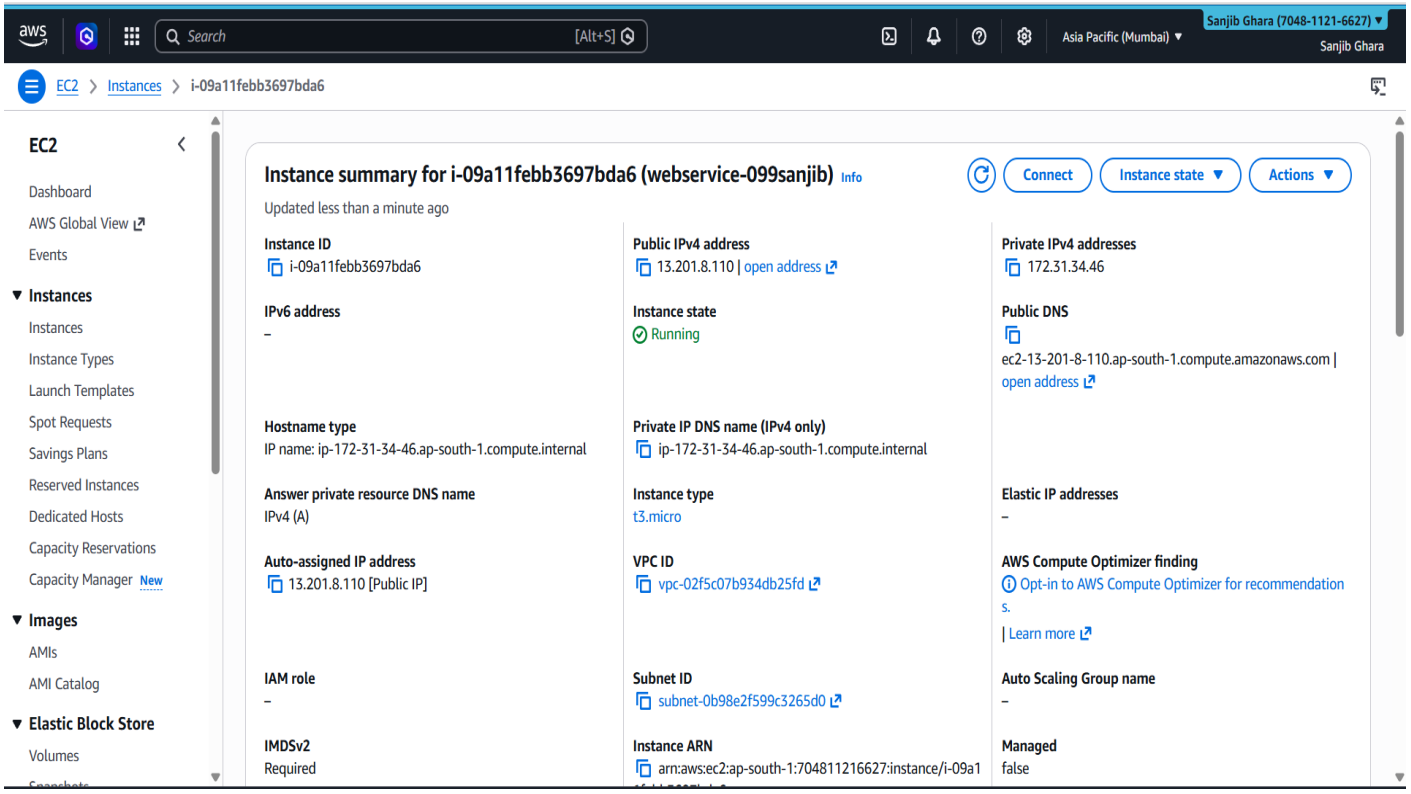
8. Download and save the key file securely.
9. Click on **Create security group** and check the boxes of **Allow HTTPS traffic from the internet and Allow HTTP traffic from the internet** and click on **Launch instance** button.



10. The Instance will be successfully created , return to the **Instance** page.



11. The Instance will be seen running, click on the newly created instance.



Step 2: Copy Public IPv4 Address

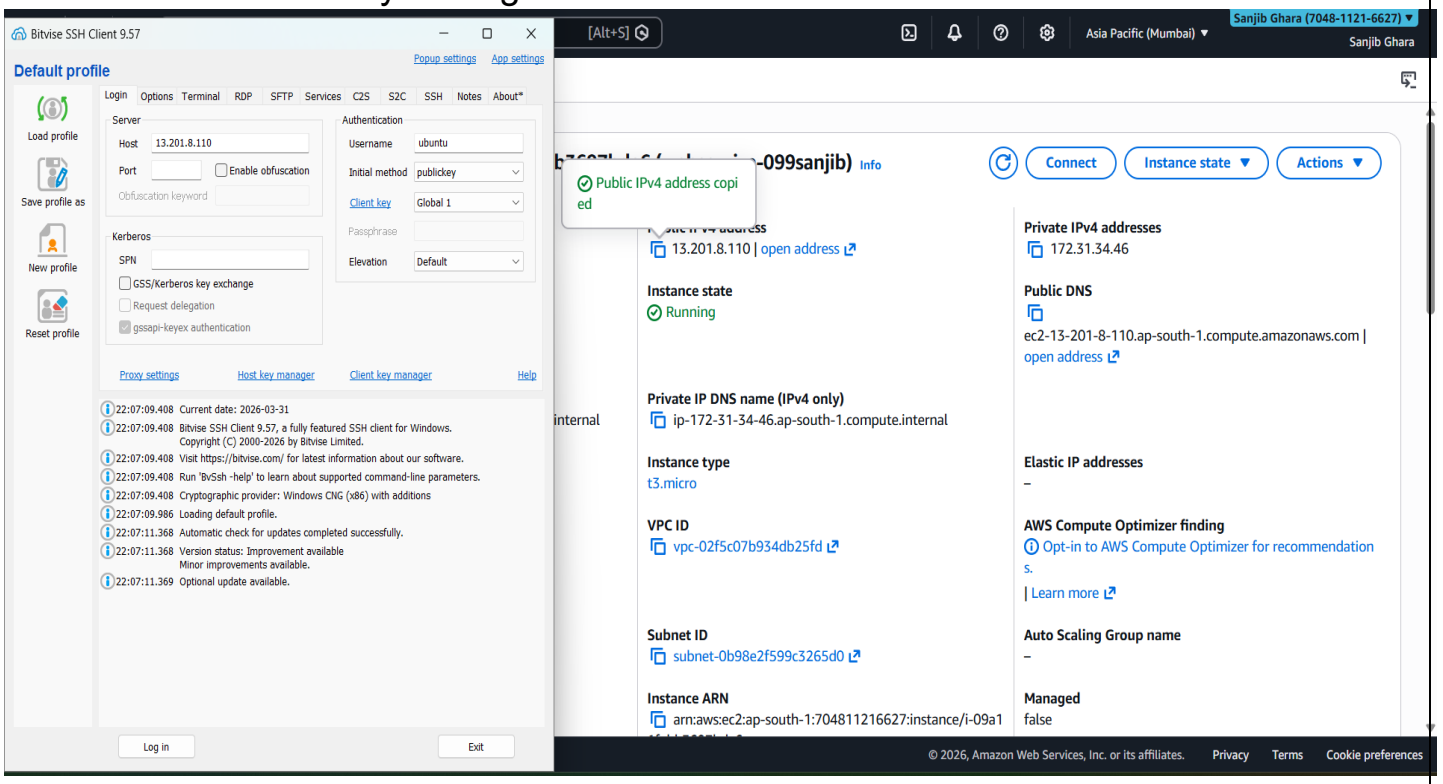
After the EC2 instance is launched and running, copy the Public IPv4 address from the Instance details page.

PHASE 2 – Connect to EC2 Instance

Step 3: Open Bitvise SSH Client and Configure Connection

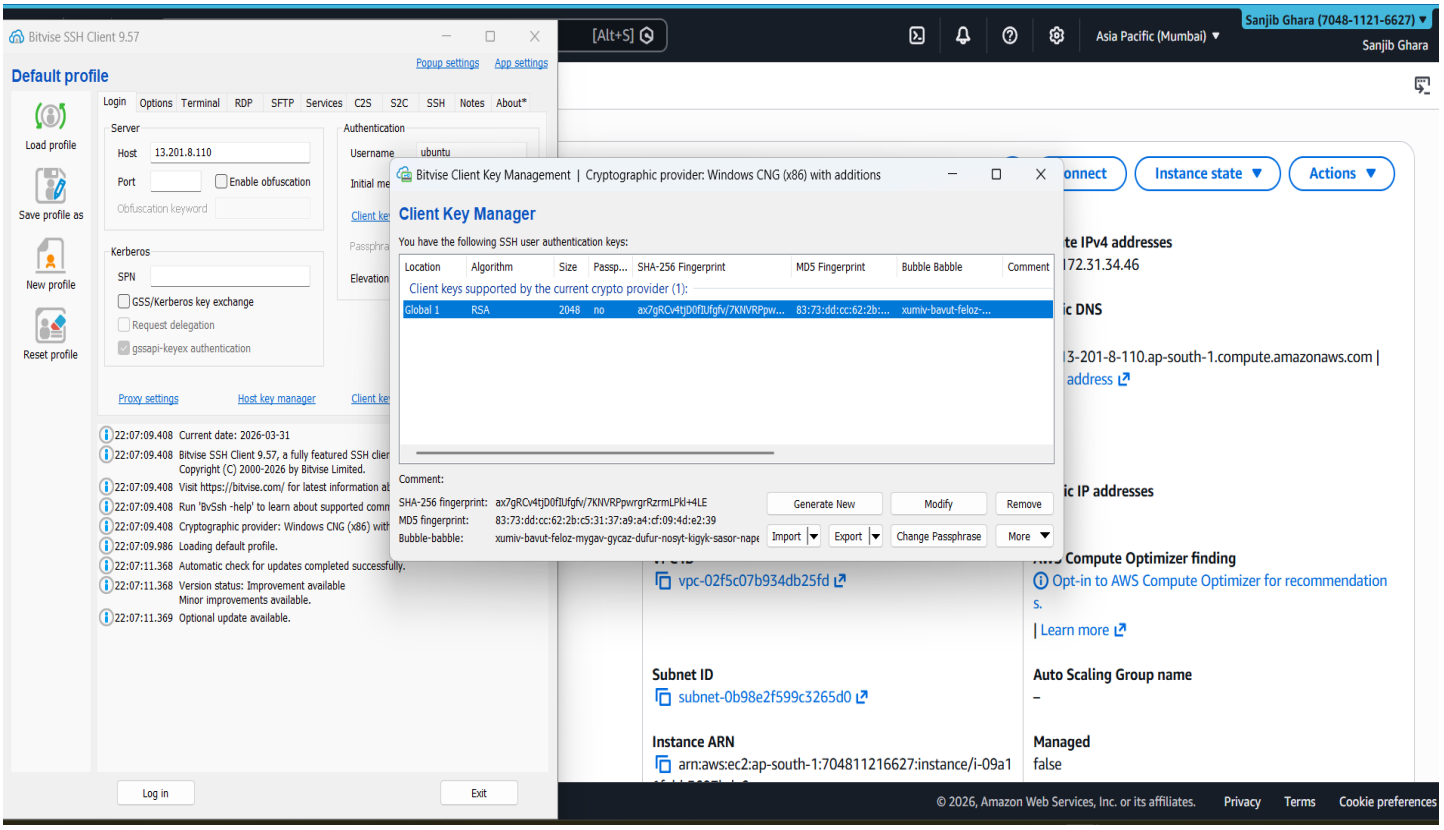
1. Open Bitvise SSH Client on the computer.
2. Paste the copied Public IPv4 address into the Host field.

3. Enter ubuntu in the Username field.
4. Set Initial method to publickey.
5. Click on Client key manager.



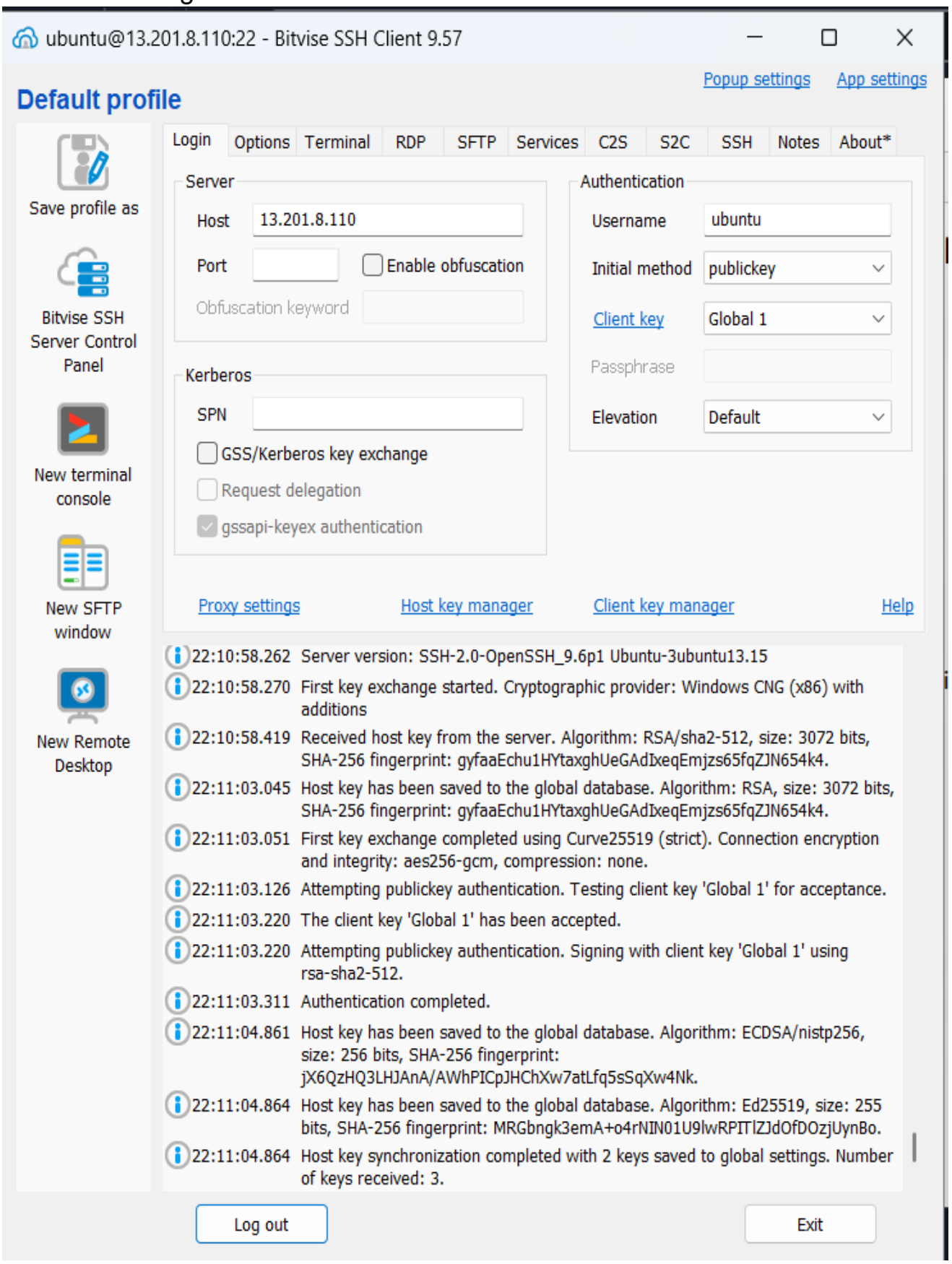
Step 4: Import Private Key

1. In Client Key Manager, click Import.
2. Select the downloaded .pem key file from the Desktop.



Step 5: Login to the Instance

1. Ensure Client key is set to Global 1.
2. Click on the Log in button.

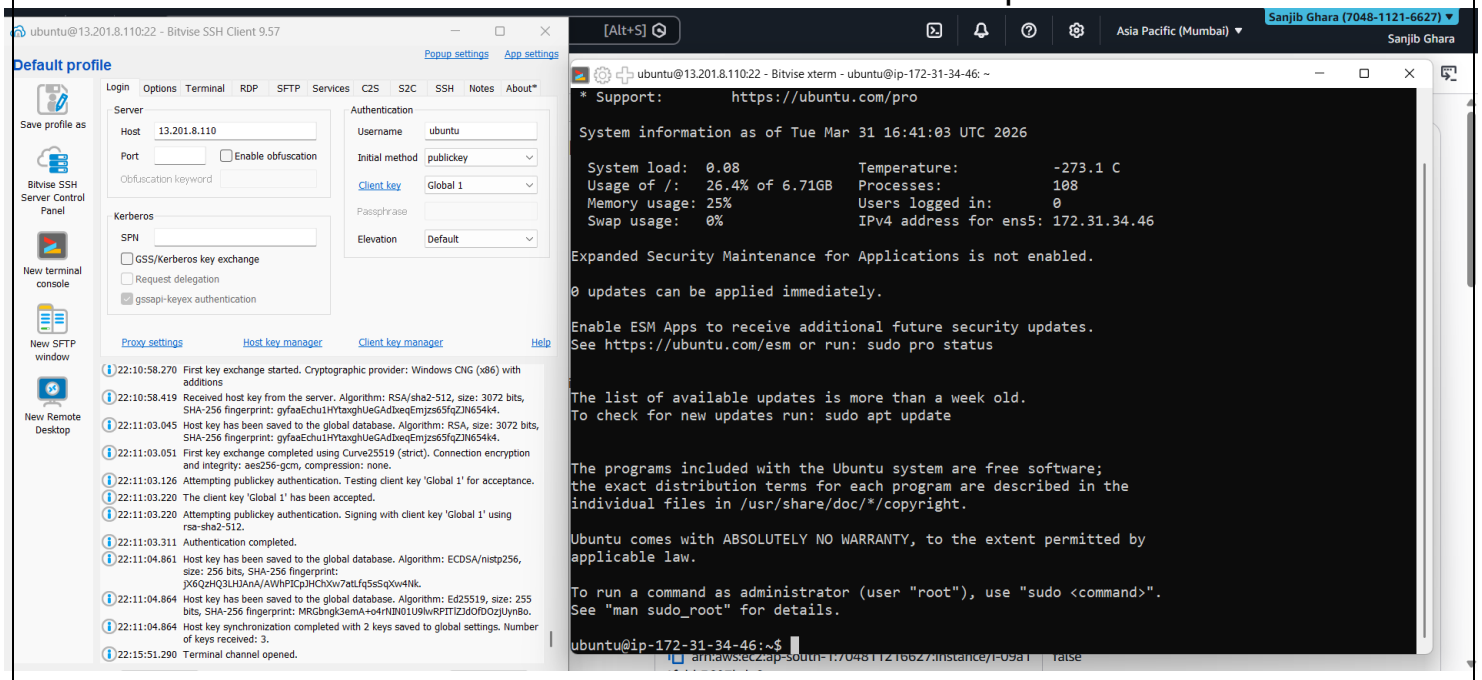


PHASE 3 – Update Linux System and Install Required Software

Step 6: Open Terminal Console

After successful login:

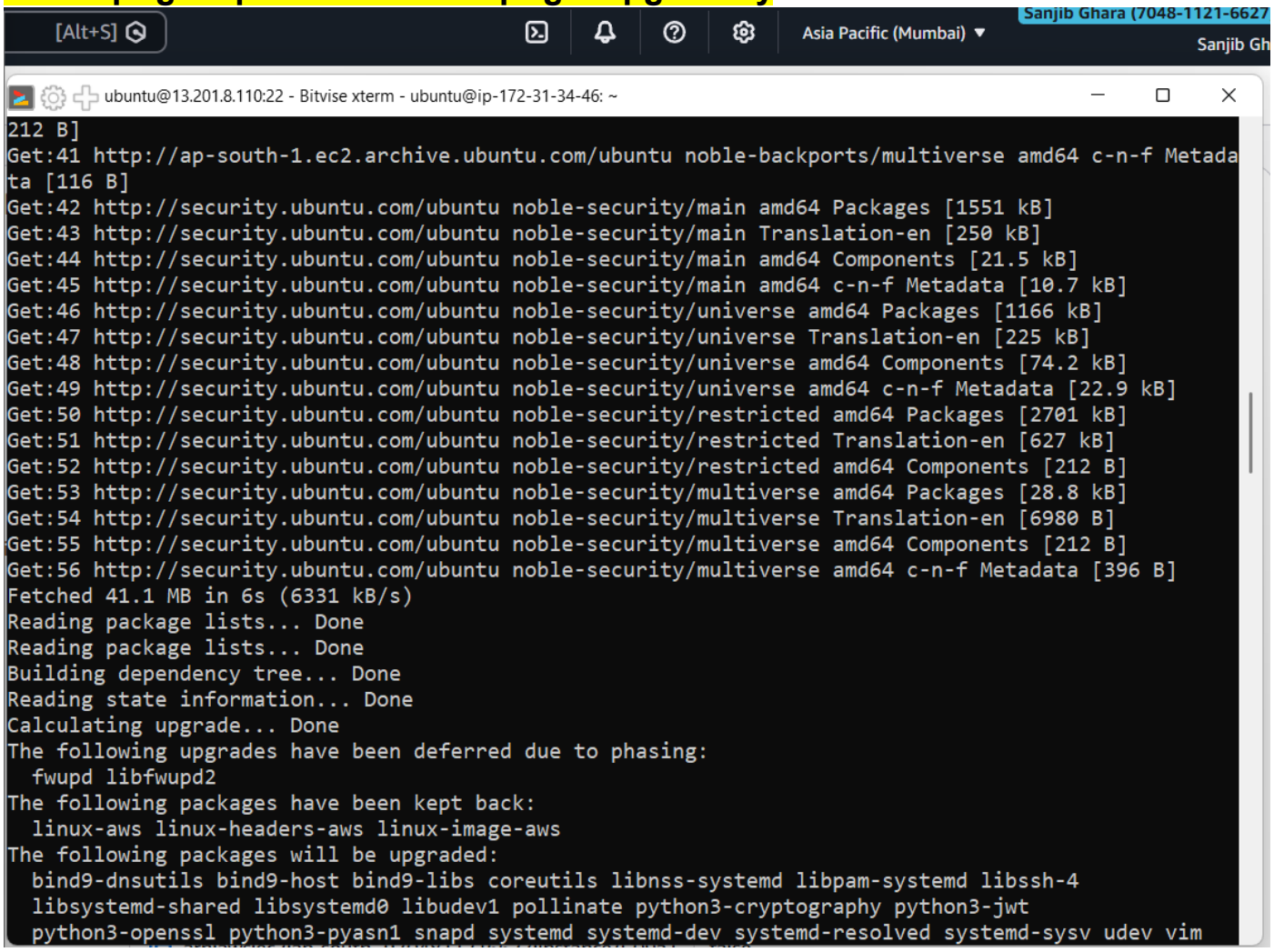
- Click on New Terminal Console from the Bitvise side panel.



Step 7: Update Ubuntu System

Run the following command to update and upgrade the system:

sudo apt-get update && sudo apt-get upgrade -y



```
ubuntu@13.201.8.110:22 - Bitvise xterm - ubuntu@ip-172-31-34-46: ~
rocessing triggers for initramfs-tools (0.142ubuntu25.8) ...
pdate-initramfs: Generating /boot/initrd.img-6.17.0-1007-aws
rocessing triggers for libc-bin (2.39-0ubuntu8.7) ...
rocessing triggers for man-db (2.12.0-4build2) ...
rocessing triggers for dbus (1.14.10-4ubuntu4.1) ...
rocessing triggers for install-info (7.1-3build2) ...
canning processes...
canning candidates...
canning linux images...

unning kernel seems to be up-to-date.

estarting services...
 systemctl restart irqbalance.service multipathd.service packagekit.service polkit.service rsyslog.s
ervice ssh.service udisks2.service

ervice restarts being deferred:
 systemctl restart ModemManager.service
 /etc/needrestart/restart.d/dbus.service
 systemctl restart networkd-dispatcher.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

o containers need to be restarted.

ser sessions running outdated binaries:
 ubuntu @ session #2: sshd[1042]
 ubuntu @ user manager service: systemd[1047]

o VM guests are running outdated hypervisor (qemu) binaries on this host.
: Some packages may have been kept back due to phasing.
buntu@ip-172-31-34-46:~$
```

Step 8: Install Nginx Web Server

Install Nginx using the command:

sudo apt-get install nginx -y

```
ubuntu@13.201.8.110:22 - Bitvise xterm - ubuntu@ip-172-31-34-46: ~
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/
nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
 * Upgrading binary nginx                                                    [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart networkd-dispatcher.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 ubuntu @ session #2: sshd[1042]
 ubuntu @ user manager service: systemd[1047]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-34-46:~$
```

Step 9: Install Node.js (LTS Version)

To install the latest supported LTS version of Node.js on Ubuntu, the NodeSource repository is configured and Node.js is installed using the following commands:

Add NodeSource Repository (Node.js 20 LTS)

curl -SL https://deb.nodesource.com/setup_16.x | sudo -E bash -

ubuntu@13.201.8.110:22 - Bitvise xterm - ubuntu@ip-172-31-34-46: ~

```
Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart networkd-dispatcher.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 ubuntu @ session #2: sshd[1042]
 ubuntu @ user manager service: systemd[1047]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 https://deb.nodesource.com/node_16.x nodistro InRelease [12.1 kB]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://deb.nodesource.com/node_16.x nodistro/main amd64 Packages [7462 B]
Fetched 19.6 kB in 1s (37.9 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
5 packages can be upgraded. Run 'apt list --upgradable' to see them.
2026-03-31 16:53:44 - Repository configured successfully. To install Node.js, run: apt install nodej
s -y
ubuntu@ip-172-31-34-46:~$
```

Install Node.js

sudo apt-get install -y nodejs

ubuntu@13.201.8.110:22 - Bitvise xterm - ubuntu@ip-172-31-34-46: ~

```
After this operation, 128 MB of additional disk space will be used.
Get:1 https://deb.nodesource.com/node_16.x nodistro/main amd64 nodejs amd64 16.20.2-1nodesource1 [27
5 MB]
Fetched 27.5 MB in 1s (39.2 MB/s)
Selecting previously unselected package nodejs.
(Reading database ... 72615 files and directories currently installed.)
Preparing to unpack .../nodejs_16.20.2-1nodesource1_amd64.deb ...
Unpacking nodejs (16.20.2-1nodesource1) ...
Setting up nodejs (16.20.2-1nodesource1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart networkd-dispatcher.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 ubuntu @ session #2: sshd[1042]
 ubuntu @ user manager service: systemd[1047]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-34-46:~$
```

Verify Installation

node -v
npm -v

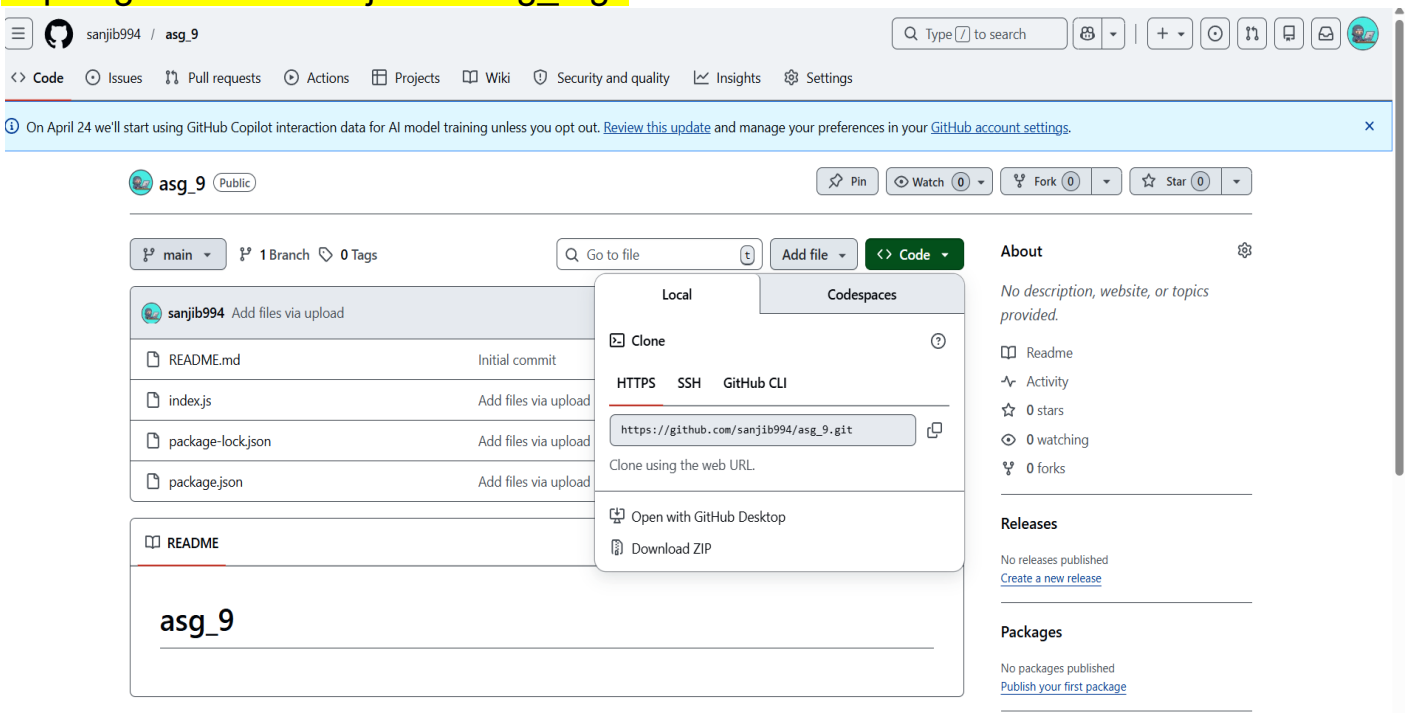
```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-34-46:~$ node -v
v16.20.2
ubuntu@ip-172-31-34-46:~$ npm -v
8.19.4
ubuntu@ip-172-31-34-46:~$
```

PHASE 4 – Clone GitHub Repository

Step 1: Copy Repository URL from GitHub

Go to GitHub → Open repository → Click Code → Copy HTTPS URL.

https://github.com/sanjib994/asg_9.git



Step 2: Clone Repository in EC2

git clone https://github.com/debatwiksantra/awslab.git

Move inside folder:

cd aws

ls

You should see:

index.js

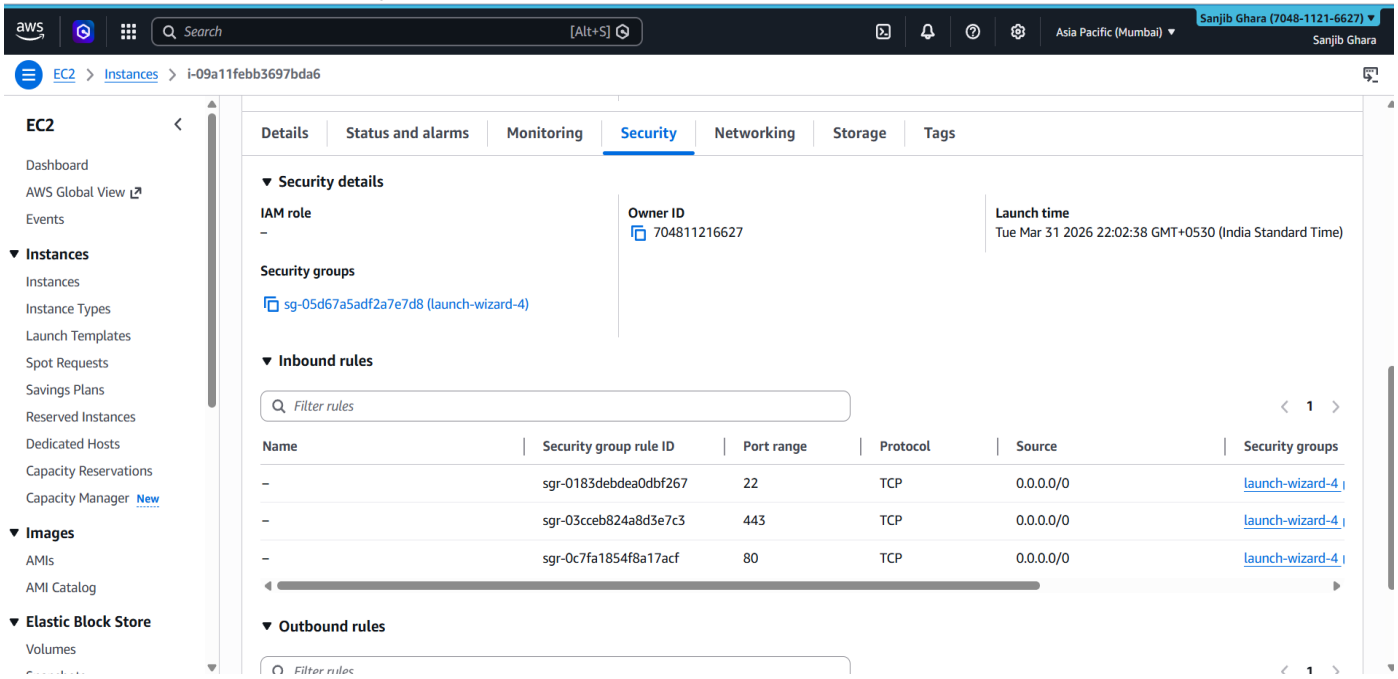
package.json

```
ubuntu@ip-172-31-34-46:~$ git clone https://github.com/sanjib994/asg_9.git
Cloning into 'asg_9'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), 49.50 KiB | 12.38 MiB/s, done.
ubuntu@ip-172-31-34-46:~$ cd asg_9
ubuntu@ip-172-31-34-46:~/asg_9$ ls
README.md  index.js  package-lock.json  package.json
ubuntu@ip-172-31-34-46:~/asg_9$
```

PHASE 5 – Configure Security Group

Step 1: Open EC2 Dashboard

1. Login to AWS Console
2. Go to EC2
3. Click Instance
4. Click on Security tab

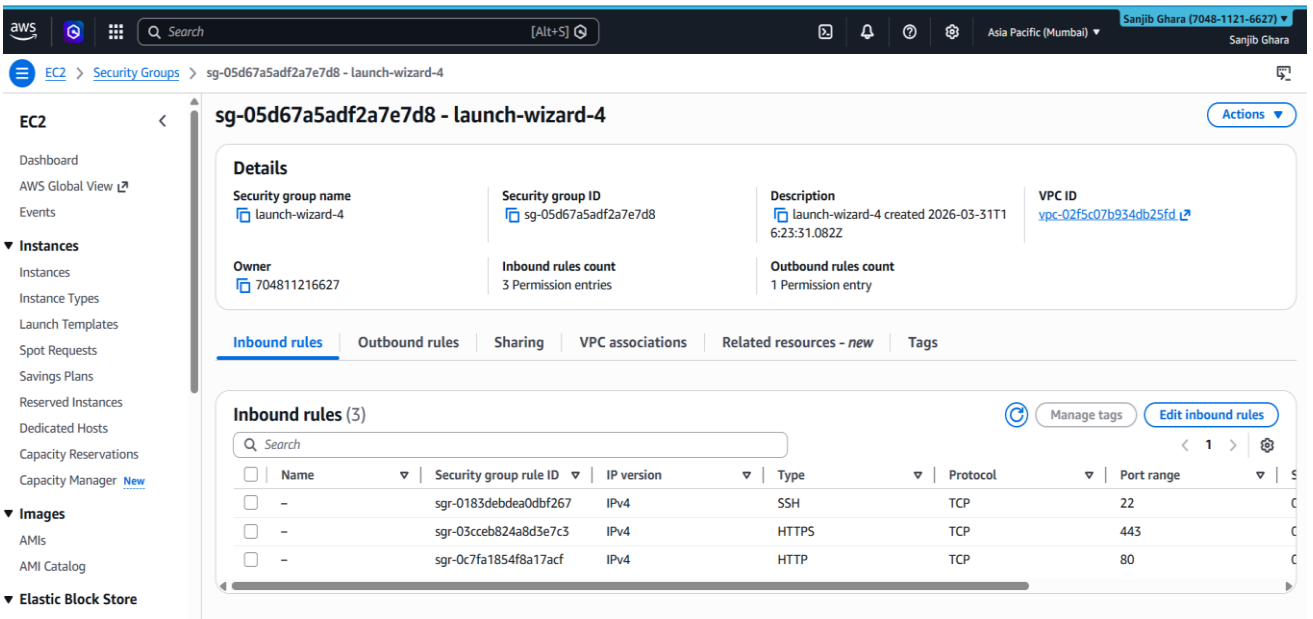


Step 2: Open Security Group

5. Under Security Groups, click on the linked Security Group name [example: sg-05d67a5adf2a7e7d8 (launch-wizard-4)] .
6. This opens the Security Group configuration page.

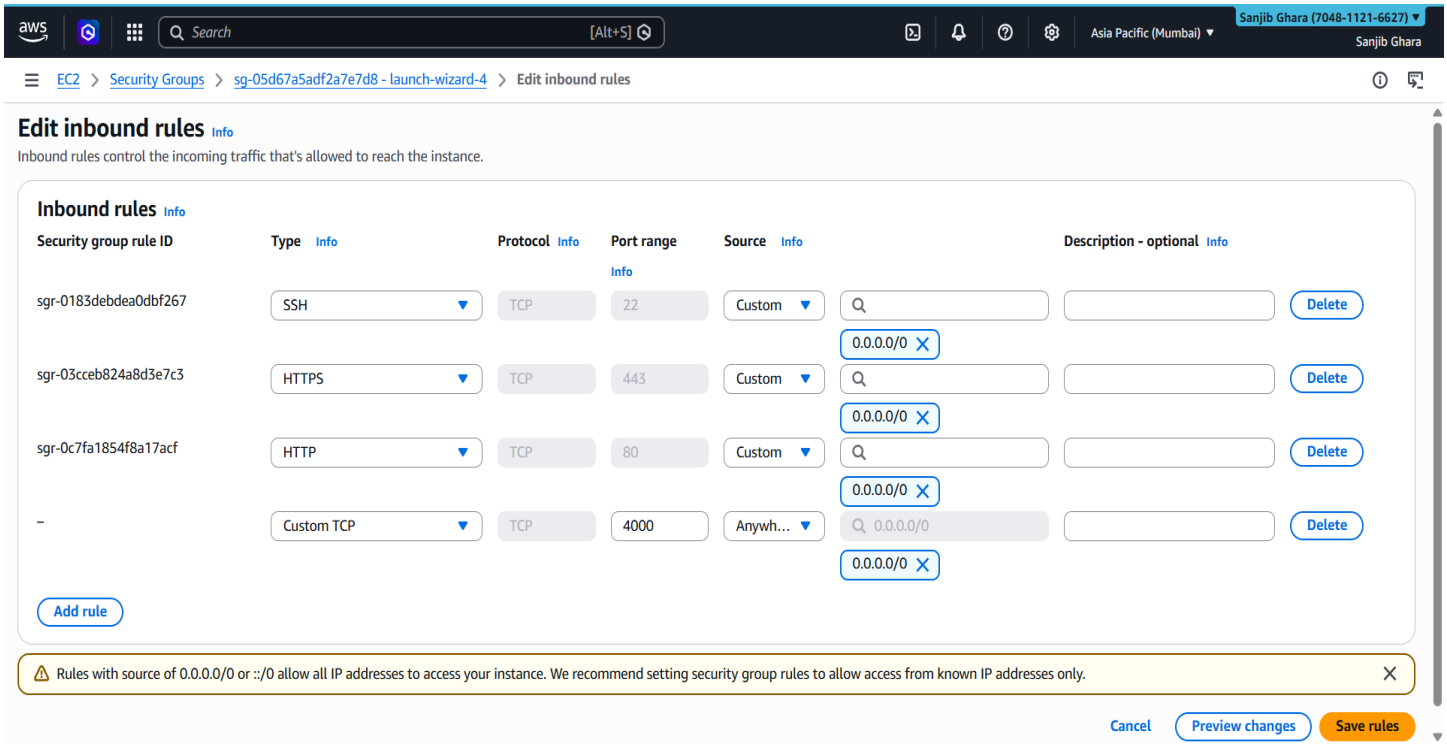
Step 3: Edit Inbound Rules

7. Click Edit inbound rules



8. Click Add Rule

Add the following rules in image:



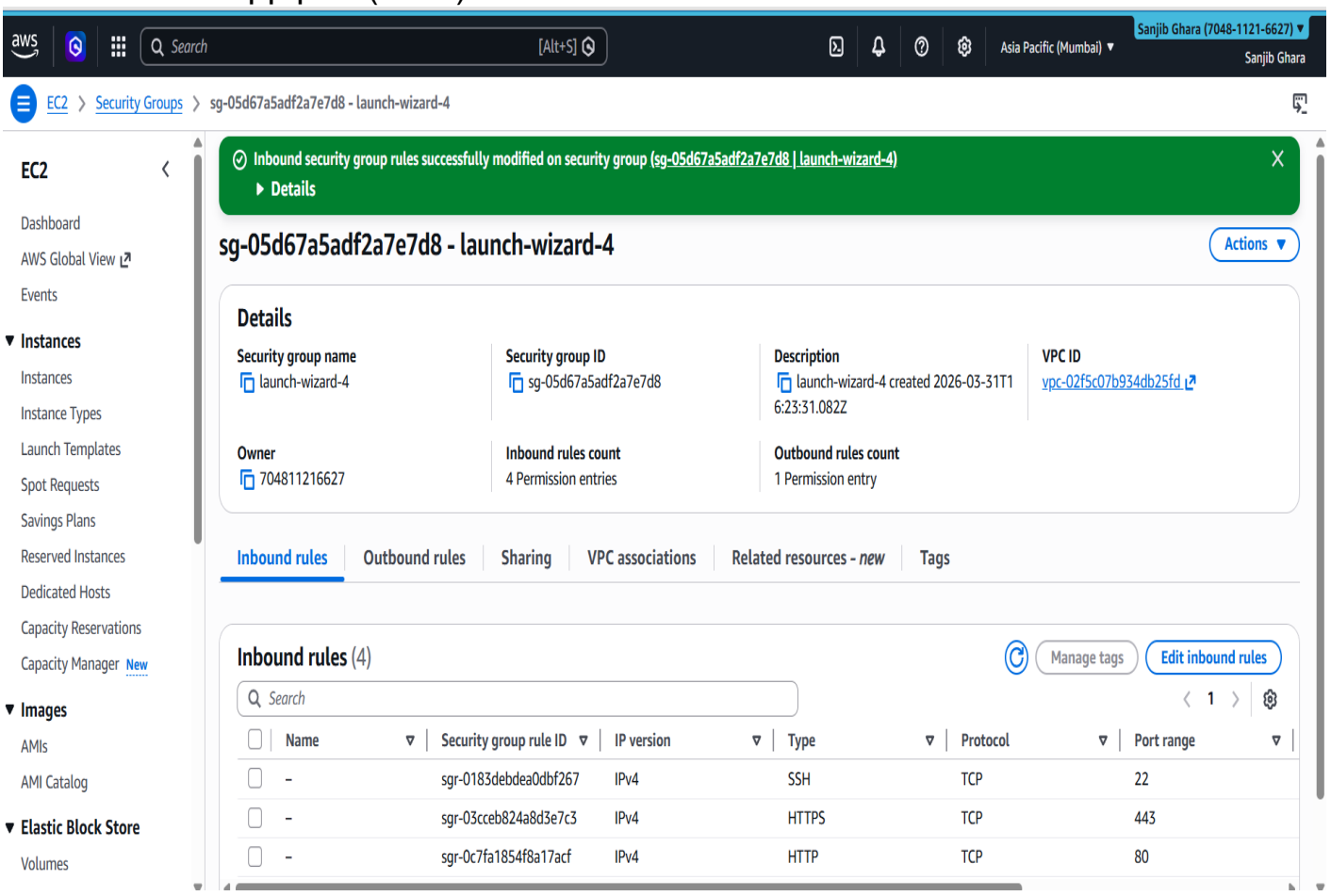
Purpose: **Allows Node.js application running on port 4000 to be accessible.**

Step 4: Save Rules

Click Save Rules.

Now the instance allows:

1. SSH connection
2. TTP access
3. HTTPS access
4. Node app port (4000)



PHASE 6 – Run the Application

Step 1: Start Application

npm install
then,
node index.jOR
npm start

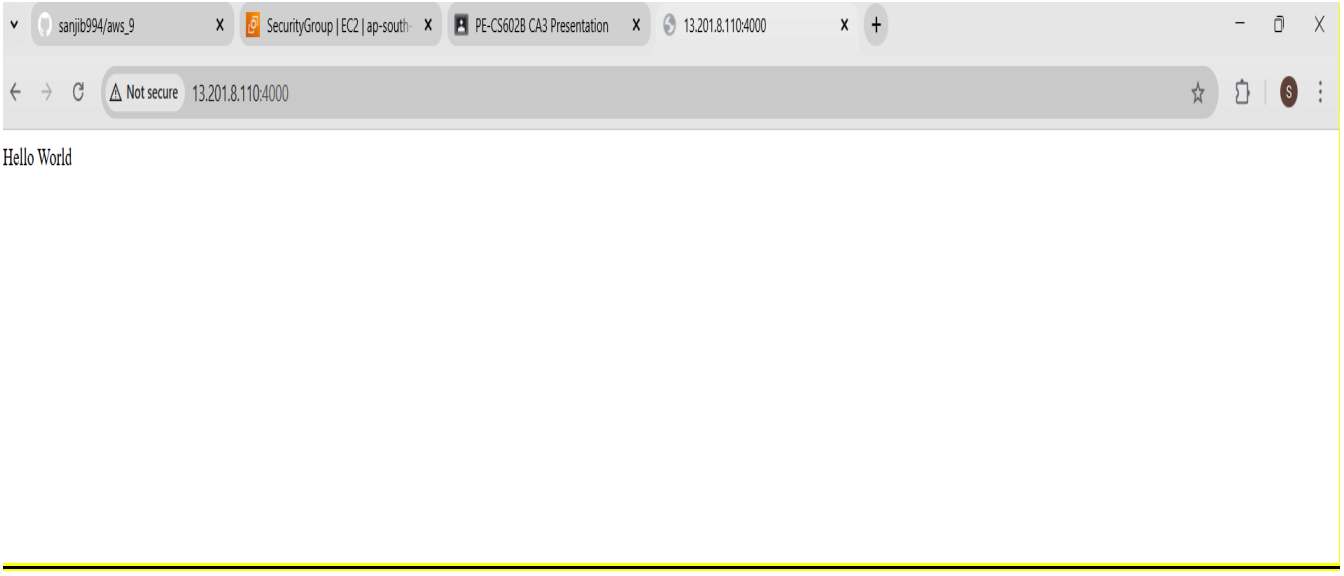
```
ubuntu@ip-172-31-34-46:~/asg_9$ node index.js
Started server
```

The static website will load successfully.

Step 2: Access Application in Browser

Open a browser and enter:
http://YOUR_PUBLIC_IP:4000

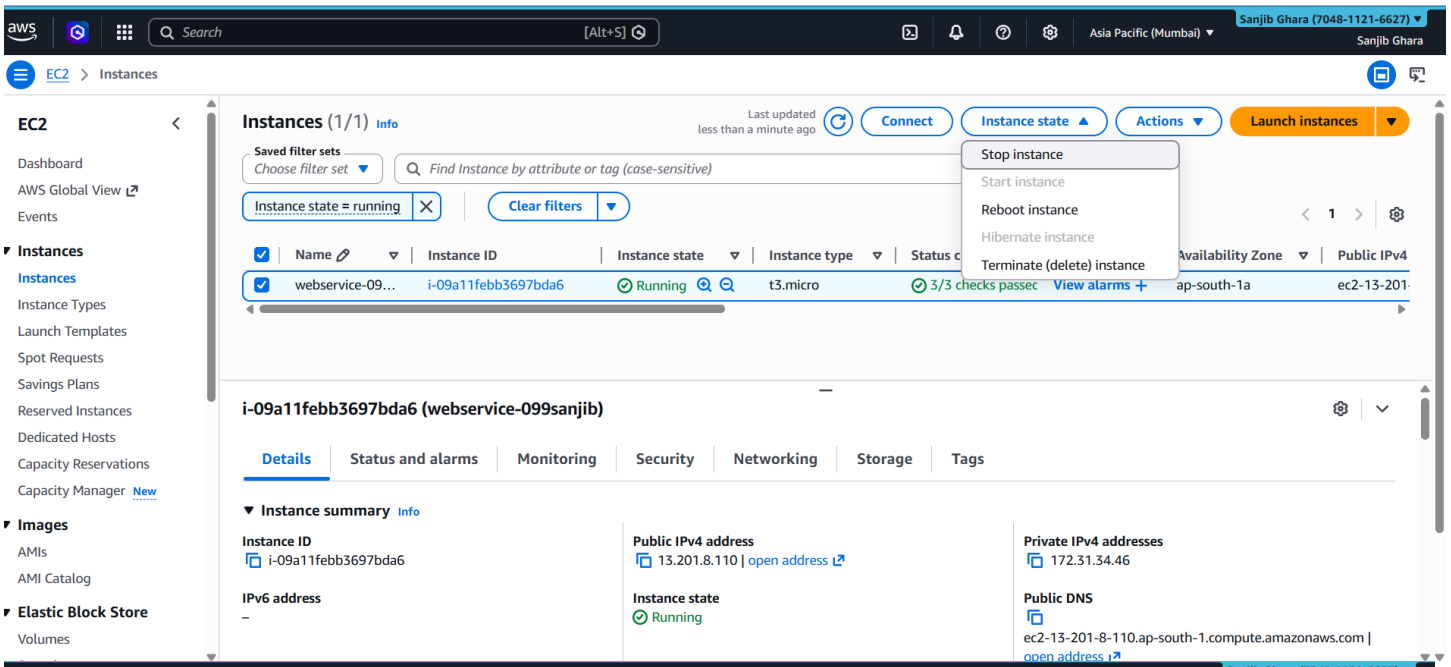
- In My Case: http://10.201.8.110:4000

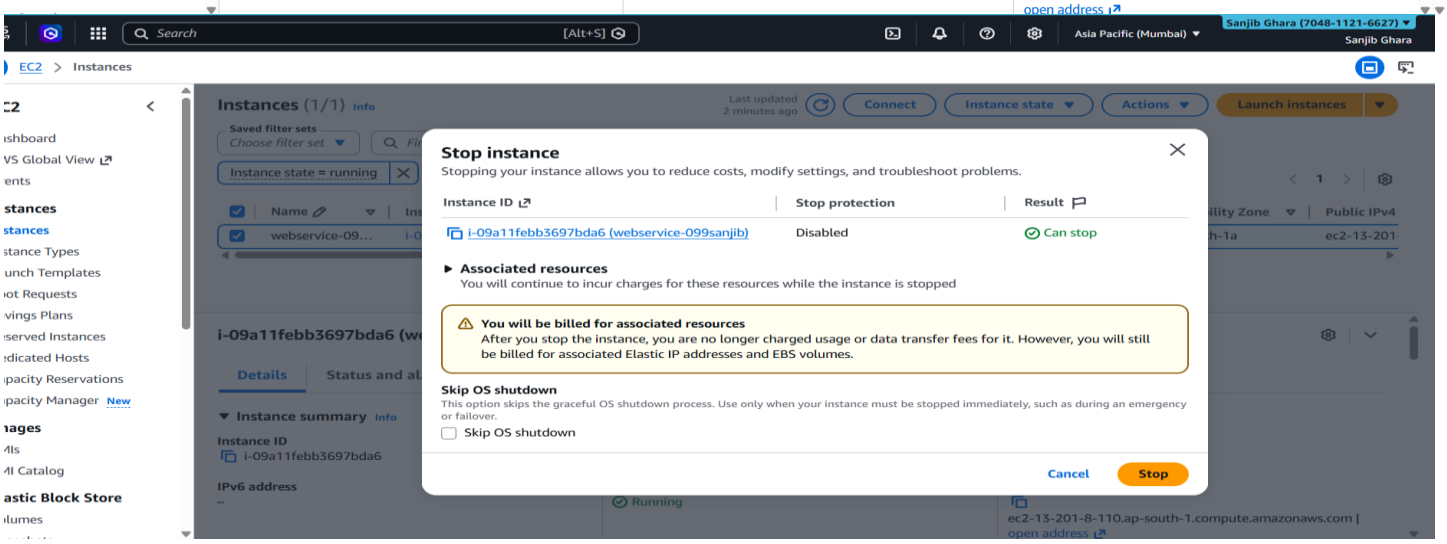


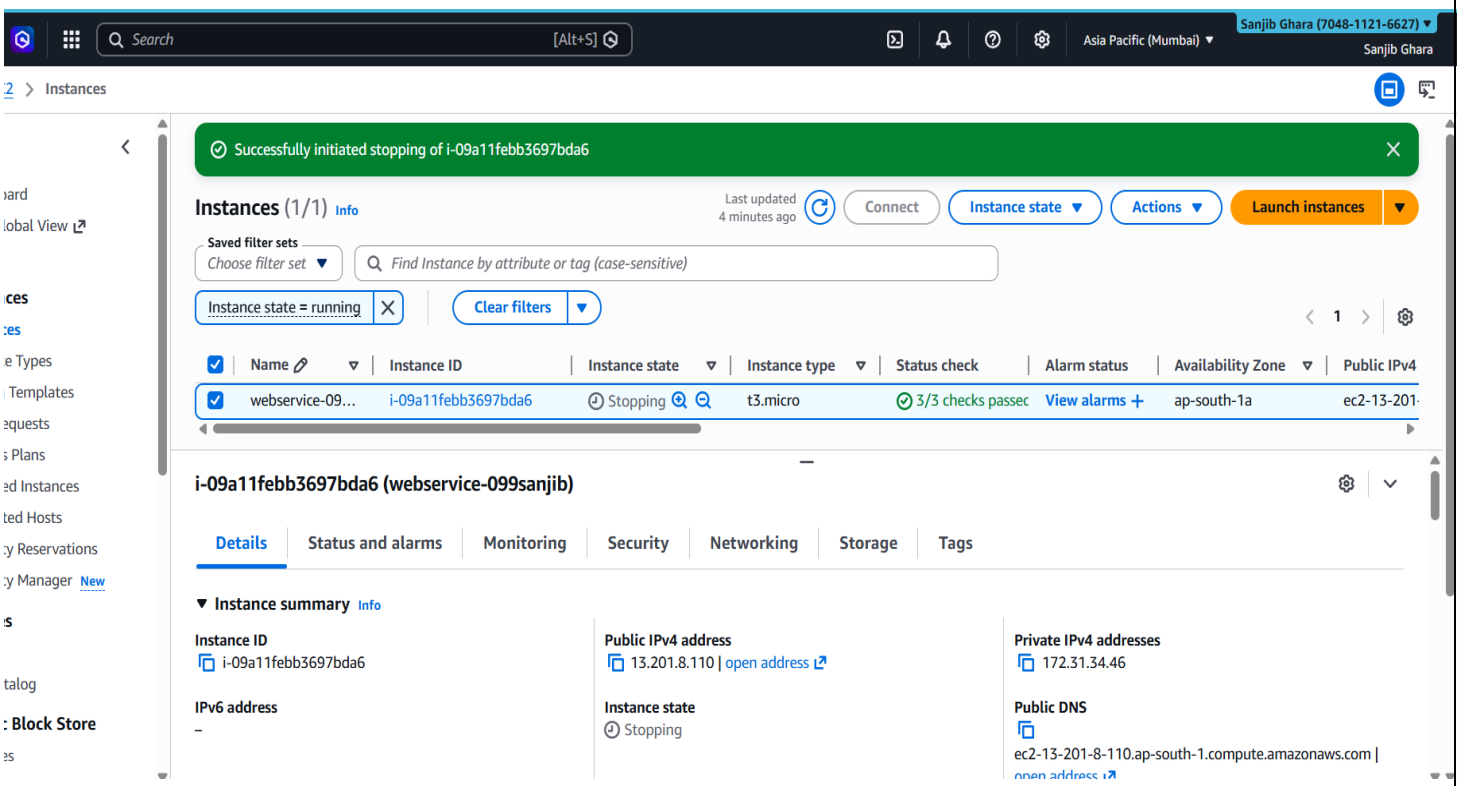
PHASE 7 – Stop and Terminate Instance

Step 1: Stop the EC2 Instance

1. Go back to the EC2 Instances page.
2. Select the instance.
3. Click Instance state → Stop instance.
4. Confirm by clicking Stop.

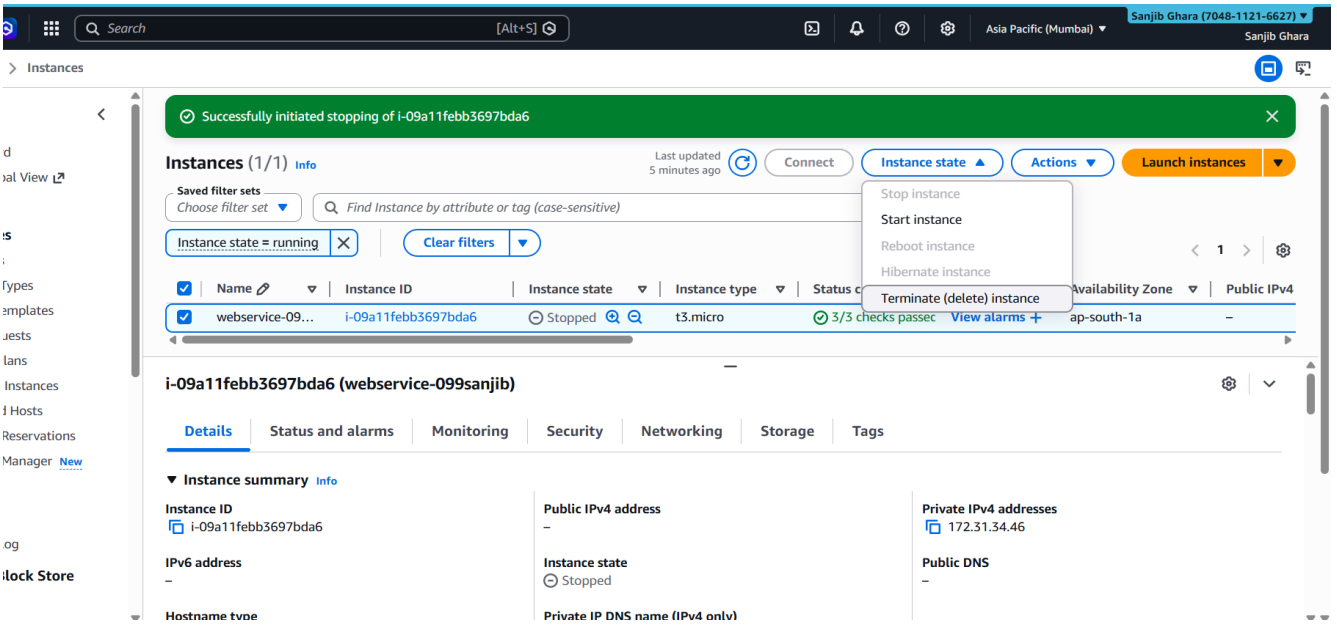

Search
[Alt+S]
Asia Pacific (Mumbai)
Sanjib Ghara (7048-1121-6627)
Sanjib Ghara
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (1/1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Stop instance
Start instance
Reboot instance
Hibernate instance
Terminate (delete) instance
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status c
Availability Zone
Public IPv4
webservice-09...
i-09a11febb3697bda6
Running
t3.micro
3/3 checks passec
View alarms +
ap-south-1a
ec2-13-201
i-09a11febb3697bda6 (webservice-099sanjib)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-09a11febb3697bda6
Public IPv4 address
13.201.8.110 | open address
Private IPv4 addresses
172.31.34.46
IPv6 address
–
Instance state
Running
Public DNS
ec2-13-201-8-110.ap-south-1.compute.amazonaws.com | open address

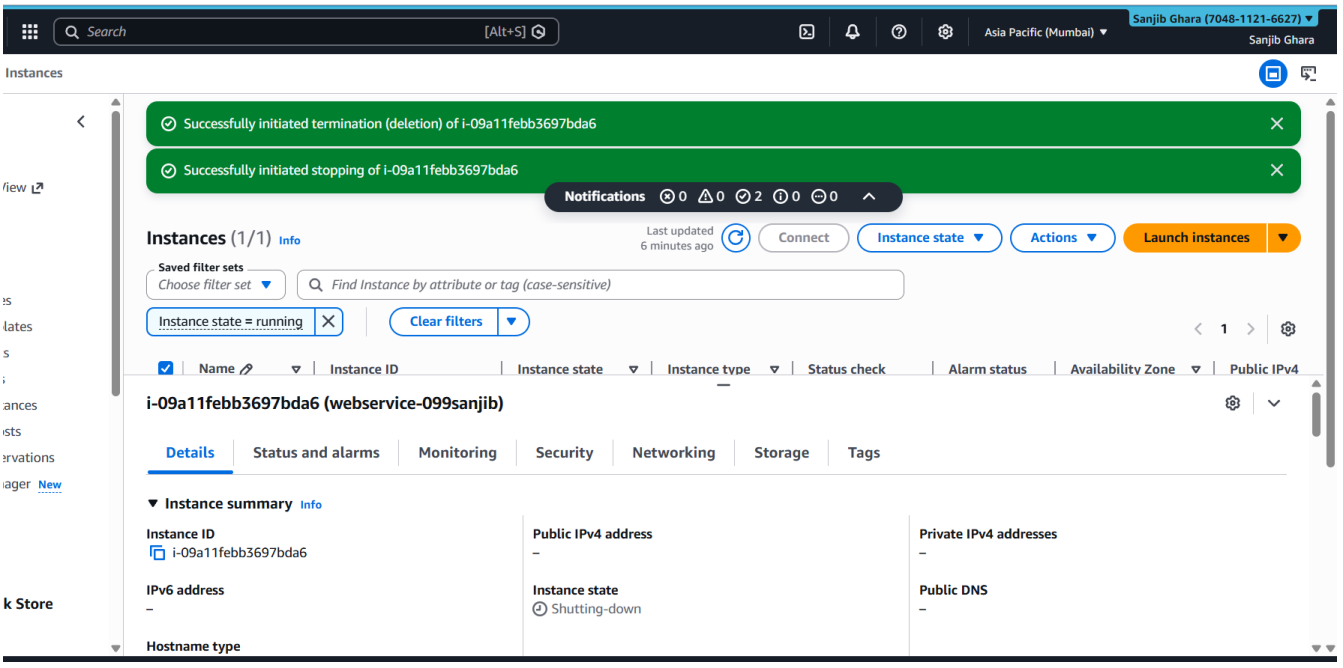

Stop instance
Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.
Instance ID
Stop protection
Result
i-09a11febb3697bda6 (webservice-099sanjib)
Disabled
Can stop
Associated resources
You will continue to incur charges for these resources while the instance is stopped
You will be billed for associated resources
After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.
Skip OS shutdown
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.
Skip OS shutdown
Cancel
Stop


Successfully initiated stopping of i-09a11febb3697bda6
Instances (1/1) Info
Last updated 4 minutes ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
webservice-09...
i-09a11febb3697bda6
Stopping
t3.micro
3/3 checks passec
View alarms +
ap-south-1a
ec2-13-201
i-09a11febb3697bda6 (webservice-099sanjib)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-09a11febb3697bda6
Public IPv4 address
13.201.8.110 | open address
Private IPv4 addresses
172.31.34.46
IPv6 address
–
Instance state
Stopping
Public DNS
ec2-13-201-8-110.ap-south-1.compute.amazonaws.com | open address

Step 2: Terminate the EC2 Instance

1. Select the same instance again.
2. Click Instance state → Terminate (delete) instance.





Conclusion

This assignment demonstrated practical understanding of:

- AWS EC2 instance management
- Linux server configuration
- Node.js deployment
- Security group configuration
- Reverse proxy setup
- Production process management